Experiment No: 3

AIM: Study of basic network command and Network configuration commands.

Date:

Competency and Practical Skills : Exploration of network commands to troubleshoot networking errors.

Relevant CO: Familiarize with the basic taxonomy - terminologies used in networking and the layered architecture of computer networks

Objectives: (a) To understand the usage of various network commands
(b) Perform commands with various options on given OS.
(c) Prepare the report with screenshots/Output and analyze the usage of each command.

Equipment/Instruments: Desktop/laptop

Theory: The operating system consists of various built-in, command-line networking utilities that are used for network troubleshooting. There are many various commands such as,

- Ping
- Ipconfig
- Tracert
- Hostname
- Pathping
- Route
- Nslookup
- Netstat
- ARP
- Getmac

1) **Ping:** Verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP) echo Request messages. The receipt of corresponding echo Reply messages are displayed, along with round-trip times. ping is the primary TCP/IP command used to troubleshoot connectivity, reachability, and name resolution. Used without parameters, this command displays Help content..

```
Command Prompt

Microsoft Windows [Version 10.0.22631.4037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hp>ping www.google.com

Pinging www.google.com [2404:6800:4009:81f::2004] with 32 bytes of data:
Reply from 2404:6800:4009:81f::2004: time=66ms
Reply from 2404:6800:4009:81f::2004: time=137ms
Reply from 2404:6800:4009:81f::2004: time=88ms
Reply from 2404:6800:4009:81f::2004: time=95ms

Ping statistics for 2404:6800:4009:81f::2004:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 66ms, Maximum = 137ms, Average = 96ms

C:\Users\hp>
```

Pic 1: Ping

2) **<u>Ipconfig</u>:** Displays all current TCP/IP network configuration values and refreshes Dynamic Host Configuration Protocol (DHCP) and Domain Name System (DNS) settings. Used without parameters, ipconfig displays Internet Protocol version 4 (IPv4) and IPv6 addresses, subnet mask, and default gateway for all adapters.

- This command is most useful on computers that are configured to obtain an IP address automatically. This enables users to determine which TCP/IP configuration values have been configured by DHCP, Automatic Private IP Addressing (APIPA), or an alternate configuration.

- If the name you supply for adapter contains any spaces, use quotation marks around the adapter name (for example, "adapter name").

- For adapter names, ipconfig supports the use of the asterisk (*) wildcard character to specify either adapters with names that begin with a specified string or adapters with names that contain a specified string. For example, Local* matches all adapters that start with the string Local and *Con* matches all adapters that contain the string Con.

- Displays the full TCP/IP configuration for all adapters. Adapters can represent physical interfaces, such as installed network adapters, or logical interfaces, such as dial-up connections.

```
C:\Users\hp>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::2c54:9cc3:be7:a8e2%7
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2401:4900:577e:ac11:5fe0:b0bb:6aff:80e0
   Temporary IPv6 Address. . . . . . : 2401:4900:577e:ac11:65:5ffc:1027:671e
   Link-local IPv6 Address . . . . . : fe80::f918:1e72:8794:2749%16
   IPv4 Address. . . . . . . . . . . : 192.168.254.51
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::8b7:e1ff:feb4:4cbc%16
                                       192.168.254.3
```

Pic 2 : Ipconfig

3) **<u>Tracert</u>:** The Windows Tracert tool determines the route to a destination by sending ICMP packets to the destination. In these packets, Tracert uses varying IP Time-To-Live values.

- The tracert command is a Command Prompt command that's used to show several details about the path that a packet takes from the computer or device you're on to whatever destination you specify.

```
C:\Users\hp>tracert www.google.com

Tracing route to www.google.com [2404:6800:4009:81f::2004]
over a maximum of 30 hops:

  1    16 ms     2 ms     2 ms  2401:4900:577e:ac11::51
  2   133 ms    39 ms    30 ms  2401:4900:20:2b::1
  3    38 ms    31 ms    44 ms  2401:4900:5a6c:34::1
  4    64 ms    29 ms    37 ms  2404:a800:2a00:20a::19
  5    90 ms    40 ms    47 ms  2404:a800::167
  6   100 ms    51 ms   164 ms  2001:4860:1:1::2d62
  7   144 ms   100 ms    55 ms  2404:6800:8100::1
  8   217 ms    51 ms    54 ms  2001:4860:0:1::4b56
  9    80 ms   147 ms    47 ms  2001:4860:0:1::4fe5
 10   143 ms    97 ms    98 ms  bom07s29-in-x04.1e100.net [2404:6800:4009:81f::2004]

Trace complete.
```

Pic 3 : Tracert

4) **<u>Hostname</u>:** This command is Displays the host name portion of the full computer name of the computer. This command is available only if the Internet Protocol (TCP/IP) protocol is installed as a component in the properties of a network adapter in Network.

```
C:\Users\hp>hostname
DEATHCORE

C:\Users\hp>
```

Pic 4 : Hostname

5) **<u>Pathping</u>:** Provides information about network latency and network loss at intermediate hops between a source and destination. This command sends multiple echo Request messages to each router between a source and destination, over a period of time, and then computes results based on the packets returned from each router. Because this command displays the degree of packet loss at any given router or link, you can determine which routers or subnets might be having network problems. Used without parameters, this command displays help.

```
C:\Users\hp>pathping ww.google.com

Tracing route to www3.l.google.com [2404:6800:4009:812::200e]
over a maximum of 30 hops:
  0  DEATHCORE [2401:4900:577e:ac11:65:5ffc:1027:671e]
  1  2401:4900:577e:ac11::51
  2  2401:4900:20:2b::1
  3  2401:4900:5a6c:34::1
  4  2404:a800:2a00:20a::19
  5  2404:a800::167
  6  2001:4860:1:1::2d62
  7  2404:6800:8114::1
  8  2001:4860:0:1::4fec
  9  2001:4860:0:1::7976
 10  2001:4860:0:1::8767
 11  2001:4860:0:1::43d9
 12  bom12s07-in-x0e.1e100.net [2404:6800:4009:812::200e]
```

Pic 5: Path ping

6) **<u>Route</u>**: The route command allows you to make manual entries into the network routing tables. The route command distinguishes between routes to hosts and routes to networks by interpreting the network address of the Destination variable, which can be specified either by symbolic name or numeric address.

```
PS C:\Users\hp> route print
===========================================================================
Interface List
  7...0a 00 27 00 00 07 ......VirtualBox Host-Only Ethernet Adapter
 22...b6 8c 9d d6 0f 21 ......Microsoft Wi-Fi Direct Virtual Adapter
 13...b6 8c 9d d6 0f 31 ......Microsoft Wi-Fi Direct Virtual Adapter #2
 16...b4 8c 9d d6 0f 71 ......MediaTek MT7921 Wi-Fi 6 802.11ax PCIe Adapter
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0    192.168.254.3   192.168.254.51     55
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
      192.168.56.0    255.255.255.0         On-link      192.168.56.1    281
      192.168.56.1  255.255.255.255         On-link      192.168.56.1    281
    192.168.56.255  255.255.255.255         On-link      192.168.56.1    281
     192.168.254.0    255.255.255.0         On-link    192.168.254.51    311
    192.168.254.51  255.255.255.255         On-link    192.168.254.51    311
  192.168.254.255  255.255.255.255         On-link    192.168.254.51    311
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link      192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link    192.168.254.51    311
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link      192.168.56.1    281
  255.255.255.255  255.255.255.255         On-link    192.168.254.51    311
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
 16     71 ::/0                     fe80::8b7:e1ff:feb4:4cbc
  1    331 ::1/128                  On-link
 16     71 2401:4900:577e:ac11::/64 On-link
 16    311 2401:4900:577e:ac11:65:5ffc:1027:671e/128
                                    On-link
 16    311 2401:4900:577e:ac11:5fe0:b0bb:6aff:80e0/128
                                    On-link
  7    281 fe80::/64                On-link
 16    311 fe80::/64                On-link
  7    281 fe80::2c54:9cc3:be7:a8e2/128
                                    On-link
 16    311 fe80::f918:1e72:8794:2749/128
                                    On-link
  1    331 ff00::/8                 On-link
  7    281 ff00::/8                 On-link
 16    311 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None
```

Pic 6 : Route

7) **<u>Nslookup</u>:** ns lookup (stands for “Name Server Lookup”) is a useful command for getting information from the DNS server. It is a network administration tool for querying the Domain Name System (DNS) to obtain domain name or IP address mapping or any other specific DNS record. It is also used to troubleshoot DNS-related problems.

```
C:\Users\hp>nslookup www.google.com
Server:  UnKnown
Address:  192.168.254.3

Non-authoritative answer:
Name:    www.google.com
Addresses:  2404:6800:4009:82c::2004
          142.251.42.36
```

Pic 7: Ns lookup

8) **Netstat:** Displays active TCP connections, ports on which the computer is listening, Ethernet statistics, the IP routing table, IPv4 statistics (for the IP, ICMP, TCP, and UDP protocols), and IPv6 statistics (for the IPv6, ICMPv6, TCP over IPv6, and UDP over IPv6 protocols). Used without parameters, this command displays active TCP connections.

```
C:\Users\hp>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49697        DEATHCORE:49698        ESTABLISHED
  TCP    127.0.0.1:49698        DEATHCORE:49697        ESTABLISHED
  TCP    127.0.0.1:49699        DEATHCORE:49700        ESTABLISHED
  TCP    127.0.0.1:49700        DEATHCORE:49699        ESTABLISHED
  TCP    127.0.0.1:49701        DEATHCORE:49702        ESTABLISHED
  TCP    127.0.0.1:49702        DEATHCORE:49701        ESTABLISHED
  TCP    192.168.254.51:53855   20.24.121.134:https    ESTABLISHED
  TCP    192.168.254.51:53976   relay-2ad8ad50:https   ESTABLISHED
  TCP    192.168.254.51:53994   20.212.88.117:https    ESTABLISHED
  TCP    192.168.254.51:54020   192.168.254.3:domain   TIME_WAIT
  TCP    192.168.254.51:54021   192.168.254.3:domain   TIME_WAIT
  TCP    192.168.254.51:54022   192.168.254.3:domain   TIME_WAIT
  TCP    192.168.254.51:54026   cdn-185-199-108-153:https  ESTABLISHED
  TCP    [2401:4900:577e:ac11:65:5ffc:1027:671e]:49409  [2603:1040:a06:6::]:https  ESTABLISHED
  TCP    [2401:4900:577e:ac11:65:5ffc:1027:671e]:53945  [2603:1046:900:2c::2]:https  ESTABLISHED
  TCP    [2401:4900:577e:ac11:65:5ffc:1027:671e]:53946  [2603:1046:900:2c::2]:https  ESTABLISHED
```

Pic 8: Netstat

9) **ARP:** Displays and modifies entries in the Address Resolution Protocol (ARP) cache. The ARP cache contains one or more tables that are used to store IP addresses and their resolved Ethernet or Token Ring physical addresses. There is a separate table for each Ethernet or Token Ring network adapter installed on your computer. Used without parameters, arp displays help information.

```
C:\Users\hp>arp -a

Interface: 192.168.56.1 --- 0x7
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
```

Pic 9: ARP

10) **Getmac:** Returns the media access control (MAC) address and list of network protocols associated with each address for all network cards in each computer, either locally or across a network. This command is particularly useful either when you want to enter the MAC address into a network analyzer, or when you need to know what protocols are currently in use on each network adapter on a computer.

```
C:\Users\hp>getmac

Physical Address    Transport Name
=================== ==========================================================
B4-8C-9D-D6-0F-71   Media disconnected
0A-00-27-00-00-07   \Device\Tcpip_{59576FD3-E5F9-42C0-A052-55AACC152191}
```

Pic 10: Get mac

Getting Help

In any command mode, you can get a list of available commands by entering a question mark (?).
Router>**?**

To obtain a list of commands that begin with a particular character sequence, type in those characters followed immediately by the question mark (?).
Router#**co?**
Configure connect copy

To list keywords or arguments, enter a question mark in place of a keyword or argument. Include a space before the question mark.
Router#**configure?**
memory Configure from NV memory
network Configure from a TFTP network
hostterminal Configure from the terminal

You can also abbreviate commands and keywords by entering just enough characters to make the commands unique from other commands. For example, youcan abbreviate the **show** command to **sh**.

Configuration Files

Any time you make changes to the router configuration, you must save the changes to memory because if you do not they will be lost if there is a system reload or power outage. There are two types of configuration files: the running (current operating) configuration and the startup configuration.
Use the following privileged mode commands to work with configuration files.

- **configure terminal** – modify the running configuration manually from the terminal.
- **show running-config** – display the running configuration.
- **show startup-config** – display the startup configuration.
- **copy running-config startup-config** – copy the running configuration to the startup configuration.
- **copy startup-config running-config** – copy the startup configuration to the running configuration.
- **erase startup-config** – erase the startup-configuration in NVRAM.

- **copy tftp running-config** – load a configuration file stored on a Trivial File Transfer Protocol (TFTP) server into the running configuration.

- **copy running-config tftp** – store the running configuration on a TFTP server.

IP Address Configuration

Take the following steps to configure the IP address of an interface.

Step 1: Enter privileged EXEC mode:
Router>**enable** password

Step 2: Enter the **configure terminal** command to enter global configuration mode.
Router#**config terminal**

Step 3: Enter the **interface** type slot/port (for Cisco 7000 series) or **interface** type port (for Cisco2500 series) to enter the interface configuration mode.
Example:
Router (config)#**interface ethernet 0/1**

Step 4: Enter the IP address and subnet mask of the interface using the **IP address** Ip address subnet mask command.Example,
Router (config-if)#**ip address 192.168.10.1 255.255.255.0**

Step 5: Exit the configuration mode by pressing Ctrl-Z
Router(config-if)#**[Ctrl-Z]**

(Give details of each command as above)

Observations: These commands are valuable for network troubleshooting, diagnosing connectivity issues, and gaining insights into the network configuration of your computer. Remember to use them responsibly and in accordance with your network administrator's policies and permissions.

Quiz: (Sufficient space to be provided for the answers)

1) **Give the sequence of troubleshoots commands you will perform with proper justification if your network is not working**

- Check Network Status on Local Device:
  Use the `ipconfig` (Windows) or `ifconfig` (Linux/macOS) command to check the IP address, subnet mask, and default gateway of your local device.

Justification: Ensure that your device has a valid IP configuration

- Ping Localhost (Loopback Test) : Run the `ping 127.0.0.1` command to check if your network stack is functional locally.

  Justification: Verifies that the network stack on your device is working.

- Check DNS Configuration : Use the `nslookup` (Windows) command to verify if DNS resolution is working properly.

  Justification : DNS issues can prevent your device from resolving domain names to IP addresses.

- Check for Network Traffic Congestion: Use tools like "netstat" or network monitoring software to identify if there is excessive network traffic or bandwidth saturation.

  Justification: Congestion can slow down or disrupt network services

Suggested Reference :

1. Microsoft: ping, https://learn.microsoft.com/en-us/windowserver/administration/windows commands/ping
2. GeeksForGeeks: Ipconfig, https://www.geeksforgeeks.org/ipconfig-full-form/
3. Educative: Tracert, https://www.educative.io/answers/what-is-the-traceroute-tracert command-and-how-is-it-used
4. IBM: Hostname, https://www.ibm.com/docs/en/cics-ts/5.3?topic=concepts-host-names
5. Microsoft: Pathping, https://learn.microsoft.com/en-us/windows server/administration/windows-commands/pathping
6. Microsoft: Route, https://learn.microsoft.com/en-us/windows server/administration/windows-commands/route_ws2008
7. Microsoft: Nslookup, https://learn.microsoft.com/en-us/windows server/administration/windows-commands/nslookup

8. JavaTPoint: Netstat, https://www.javatpoint.com/linux-netstat
9. Wikipedia: ARP, https://en.wikipedia.org/wiki/Address_Resolution_Protocol
10. Microsoft: Getmac, https://en.wikipedia.org/wiki/Coaxial_cable https://learn.microsoft.com/en-us/windows-server/administration/windows commands/getmac

Rubric wise marks obtained:

	3	2	1
Ability to identify and connect networking cables	Clear (Good)	Average/partial	Poor/not at all
Organization	Can easily tell from which sources information was drawn	Can tell with difficulty from where information came	Cannot tell from which source information came
Bibliography	All relevant bibliographic information is included	Bibliography contains most relevant information	Bibliography contains very little information
cross-wired cable and straight through cable using clamping tool	Cable working properly with all connections looks proper	Cable working properly with all connections looks poorly connected	Cable not working properly

Experiment No: 4

AIM: Implement different LAN topologies using Network Simulator.

Date:

Competency and Practical Skills: Exploration of network layout to connect devices with each other to establish network

Relevant CO: CO-5 Implement different types of network using different tools and simulators

Objectives:
(a) Download network simulator – preferably cisco packet tracer
(b) Implement topologies by configuring interconnecting devices and computing devices.
(c) Analys the performance of connecting devices in different ways

Equipment/Instruments: Desktop/laptop, Network simulator

Theory:

What is Network Topology?

The arrangement of a network that comprises nodes and connecting lines via sender and receiver is referred to as network topology.

There are various types of topologies and they are:

- Bus Topology
- Star Topology
- Ring Topology
- Mesh Topology
- Hybrid Topology

1. **<u>Bus Topology</u>:** Bus Topology is a setup of a network where network devices, computers or nodes are directly linked to a single cable.

Bus topology using Packet Tracer:

- Take 4 switches, connect them through single cable.
- Each of the switches will be connected with an end device whose IP addresses are given.
- To configure the IP addresses of the devices "Click on the device for which you want to configure the IP address-> Click Desktop -> IP configuration -> Provide IP of your choice -> Press Tab for automatically having Subnet Mask which is: 255.0.0.0 for all the devices available in the topology.

- Provide IP addresses to all the End devices only and then you can send the data among these devices.

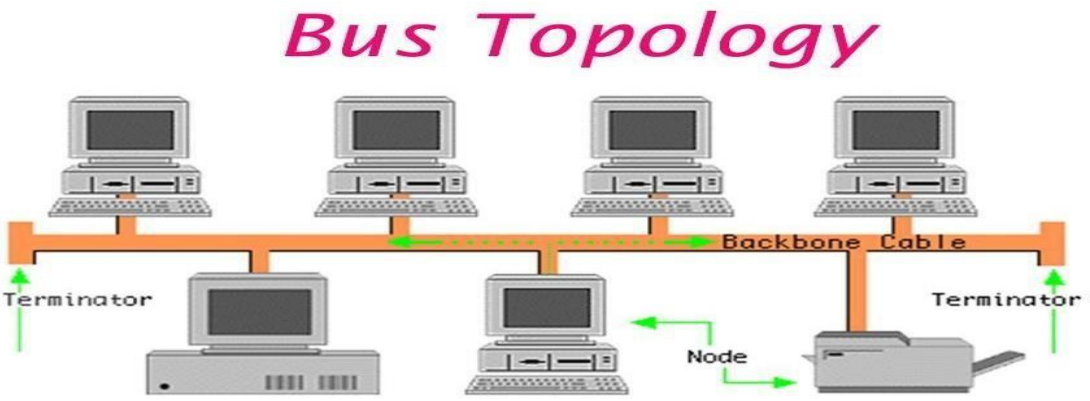


Fig 1.2 : Bus Topology

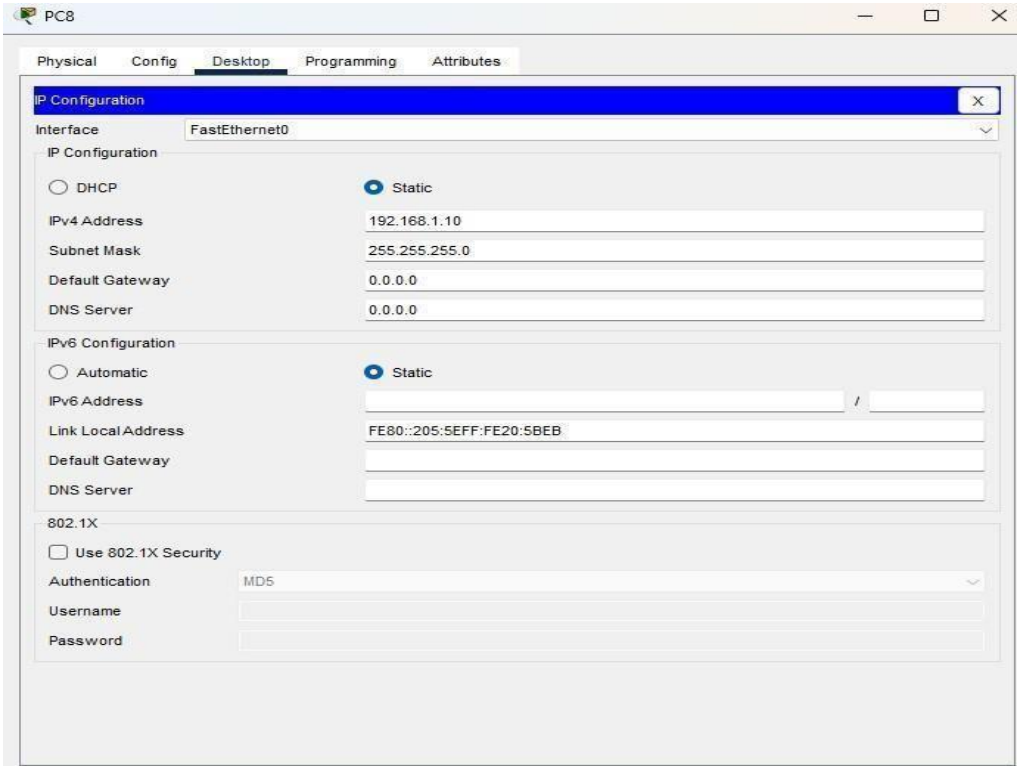


Fig 1.2: Static IP Configuration

- Check the network by sending messages/packets from one device to another and check the status of the message as “Successful”. If the messages are not sent due to some reasons it will show “Failed” as the status of the message.

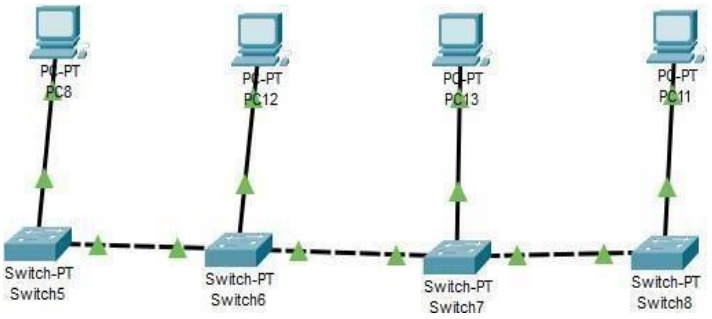


Fig 1.3: Bus topology configuration in simulator

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC8	ICMP
	0.001	PC8	Switch5	ICMP
	0.002	Switch5	Switch6	ICMP
	0.003	Switch6	Switch7	ICMP
	0.004	Switch7	Switch8	ICMP
	0.005	Switch8	PC11	ICMP
	0.006	PC11	Switch8	ICMP
	0.007	Switch8	Switch7	ICMP
	0.008	Switch7	Switch6	ICMP
	0.009	Switch6	Switch5	ICMP
Visible	0.010	Switch5	PC8	ICMP

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)	

Fig 1.4: Bus topology simulation

2. <u>Star Topology</u>: A star topology is a topology for a Local Area Network (LAN) in which all nodes are individually connected to a central connection point, sort of a hub or a switch.

Star Topology Using Packet Tracer:

- Have a single switch which will be the central node.
- All the end devices are connected with the central switch so if any of the cable fail occurs it does not affect the whole system.
- Configure the IP address of all the connected devices.
- Send messages to other devices connected to the same Switch.
- As we can see in the below image we can send the messages from one device to another and we can also see the status of the message as "Successful" here. If the messages are not sent due to some reasons it will show "Failed" as the status of the message.

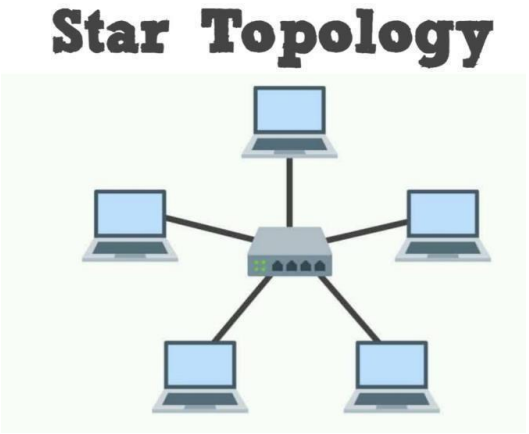


Fig 2.1: Star Topology

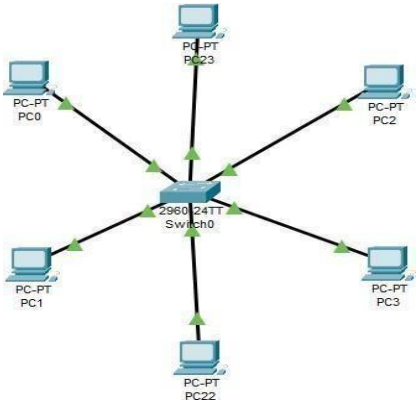


Fig 2.2: Star Topology in Simulator

- Check the network by sending messages/packets from one device to another and check the status of the message as “Successful”. If the messages are not sent due to some reasons it will show “Failed” as the status of the message.

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.002	Switch0	PC3	ICMP
	0.003	PC3	Switch0	ICMP
Visible	0.004	Switch0	PC0	ICMP

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)	

Fig 2.3: Star Topology in Simulation

3. Ring Topology:

- In this topology, it forms a ring connecting devices with exactly two neighboring devices.
- It is a network configuration where device connections create a circular data path. In this each device is connected to with its exactly two neighboring devices, like points on a circle which forms like a ring structure.
- The most common access method of ring topology is token passing.

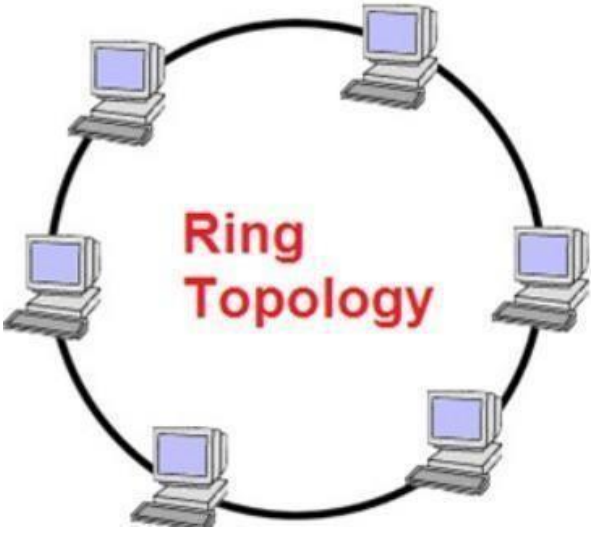


Fig 3.1: Ring Topology

Ring Topology using Packet Tracer:

- Take 4 switches and connect them with each other.
- Each of the hubs is connected with an end device whose IP addresses can be configured
- After configuring the IP addresses send messages to other devices connected in the Network.
- As we can see in the below image we can send the messages from one device to another and also see the status of the message as "Successful" here. If the messages are not sent due to some reasons it will show "Failed" as the status of the message.

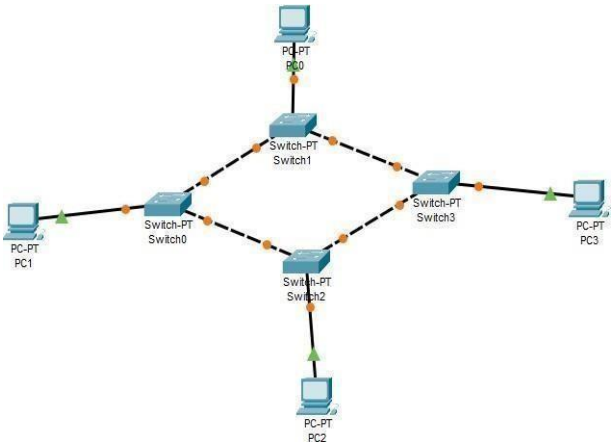
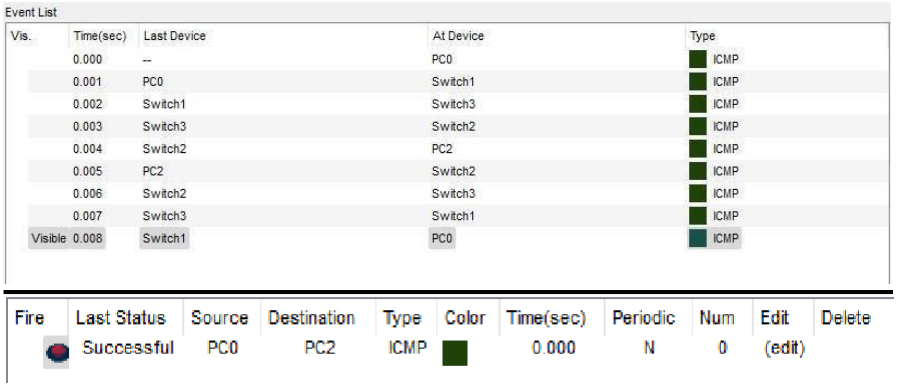


Fig 3.2: Ring Topology in Simulator

- Check the network by sending messages/packets from one device to another and check the status of the message as "Successful". If the messages are not sent due to some reasons it will show "Failed" as the status of the message.



Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Switch1	ICMP
	0.002	Switch1	Switch3	ICMP
	0.003	Switch3	Switch2	ICMP
	0.004	Switch2	PC2	ICMP
	0.005	PC2	Switch2	ICMP
	0.006	Switch2	Switch3	ICMP
	0.007	Switch3	Switch1	ICMP
Visible	0.008	Switch1	PC0	ICMP

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)	

Fig 3.3: Ring Topology in Simulation

4. <u>Mesh Topology</u>: In a mesh topology, every device is connected to another device via a particular channel.

- There are two types of Mesh topologies:
  1. Fully-connected Mesh Topology
  2. Partially-connected Mesh Topology

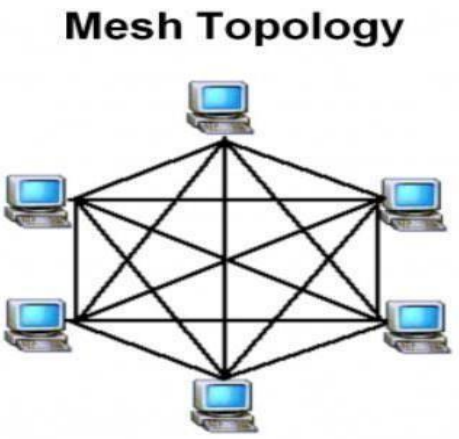


Fig 4.1 : Mesh Topology

Mesh Topology using Packet Tracer:

- 4 network switches which are connected to each other through network cables. Each of the switch is connected all the other switches as well as an end device.
- Configure the end devices and then can send the message to any other device connected to particular network.
- As we can see in the below image we can send the messages from one device to another and we can also see the status of the message as "Successful" here. If the messages are not sent due to some reasons it will show "Failed" as the status of the message.

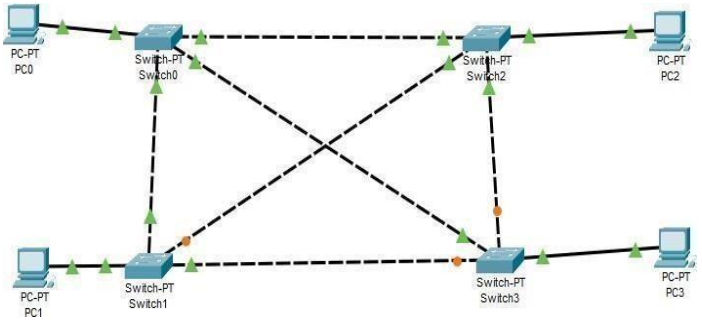


Fig 4.2: Mesh Topology in Simulator

- Check the network by sending messages/packets from one device to another and check the status of the message as “Successful”. If the messages are not sent due to some reasons it will show “Failed” as the status of the message.

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.002	Switch0	Switch3	ICMP
	0.003	Switch3	PC3	ICMP
	0.004	PC3	Switch3	ICMP
	0.005	Switch3	Switch0	ICMP
	0.006	Switch0	PC0	ICMP
	0.713	--	Switch0	STP
	0.714	Switch0	PC0	STP
	0.714	Switch0	Switch2	STP
	0.714	Switch0	Switch1	STP
	0.714	Switch0	Switch3	STP
Visible	0.715	Switch2	PC2	STP
Visible	0.715	Switch2	Switch3	STP
Visible	0.715	Switch2	Switch1	STP
Visible	0.715	Switch1	PC1	STP
Visible	0.715	Switch1	Switch3	STP
Visible	0.715	Switch3	PC3	STP

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	

Fig 4.3: Mesh Topology in Simulation

5. **<u>Hybrid Topology</u>:** A network structure whose design contains more than one topology is said to be hybrid topology. Hybrid topology inherits merits and demerits of all the incorporating topologies.

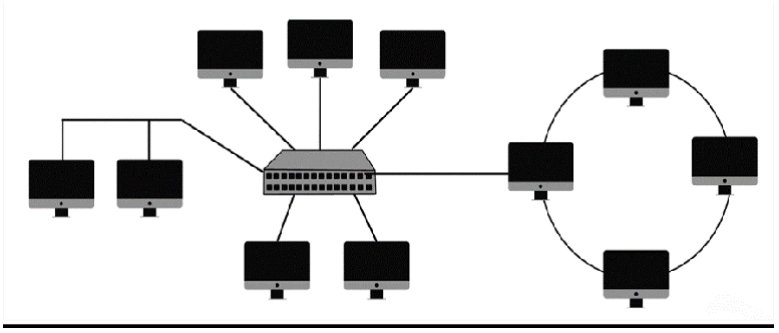


Fig 5.1: Hybrid Topology

Hybrid Topology Using Packet Tracer:

- Have 4 network switches which are connected to each other through network cables forming ring.
- Then have central switch which is connected to one of the switches used in forming ring and also to other devices forming star, thus forming hybrid topology.
- Each of the switch is connected all the other switches as well as an end device.
- We can configure the end devices as seen before and then can send the message to any other device connected to particular network.

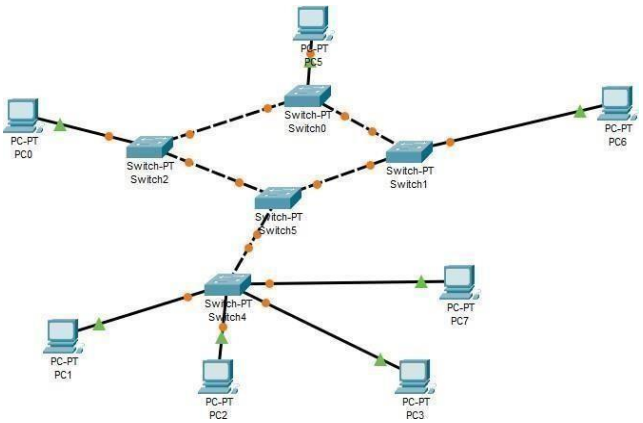


Fig 5.2: Hybrid Topology in Simulator

- Check the network by sending messages/packets from one device to another and check the status of the message as "Successful". If the messages are not sent due to some reasons it will show "Failed" as the status of the message.

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC5	ICMP
	0.001	PC5	Switch0	ICMP
	0.002	Switch0	Switch2	ICMP
	0.003	Switch2	Switch5	ICMP
	0.004	Switch5	Switch4	ICMP
	0.005	Switch4	PC2	ICMP
	0.006	PC2	Switch4	ICMP
	0.007	Switch4	Switch5	ICMP
	0.008	Switch5	Switch2	ICMP
	0.009	Switch2	Switch0	ICMP
Visible	0.010	Switch0	PC5	ICMP

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC5	PC2	ICMP		0.000	N	0	(edit)	

Fig 5.3: Hybrid Topology in Simulation

Observations:

(1) Which topology or topologies are implemented at your institute?

Our institute implemented Star Topology.

(2) If there is a small setup of your office having 10 Computers connected, which topology you will prefer? Why?

I prefer star topology because, In a star topology, all computers are connected to a central hub or switch. If one computer or cable fails, it doesn't affect the rest of the network. This topology is easy to set up, manage, and troubleshoot, making it a good choice for small offices without dedicated IT staff and It's relatively easy to add more devices to the network in the future.

(3) Interpret and analyze the output of each topology.

1. **Bus Topology:** Bus topology can be efficient for small to medium-sized networks with a limited number of devices.

2. **Star Topology:** Network management and troubleshooting are relatively straightforward in a star topology because all devices are connected to a central point. It's easier to monitor and diagnose network issues.
3. **Ring Topology:** Analyzing the output of a ring topology involves a combination of monitoring tools, network management practices, and a deep understanding of the network's design and requirements. Regularly assessing the network's performance and health is essential to ensure it meets the needs of its users and remains reliable.

4. **Mesh Topology:** As the network grows, the benefits of redundancy and reliability become more pronounced. However, the complexity of managing the network also increases. Organizations must carefully consider their specific needs and resources when deciding to implement a mesh topology.

5. **Hybrid Topology:** analyzing the output of a hybrid network topology involves a comprehensive assessment of its configuration, performance, reliability, security, and scalability. This analysis should help you identify strengths and weaknesses, make informed decisions for network optimization, and ensure that the network meets the organization's requirements effectively.

Quiz:

(1) Give Advantages, disadvantages of each topology.

1. Bus Topology:

- **Advantages:**
  - It is easy to set up, handle, and implement.
  - It costs very less.
- **Disadvantages:**
  - cable length is limited. This limits the number of network nodes that can be connected.

2. Star Topology:

- **Advantages:**
  - It achieves isolation of each device in the network.
  - Adding or removing network nodes is easy. ➢
- **Disadvantages:**
  - The setup cost is quite high.

3. Ring Topology:

- **Advantages:**
  - traffic is unidirectional and the data transmission is high-speed.
  - The configuration makes it easy to identify faults in network nodes.
- **Disadvantages:**
  - There is heavy dependency on the wire connecting the network nodes in the ring.

4. Mesh Topology:

- **Advantage:**
  - Point-to-point contact between every pair of nodes, makes it easy to identify faults.
- **Disadvantages:**
  - lot of cabling is required. Thus, the costs incurred in setup and maintenance are high.

5. Hybrid Topology:

- **Advantages:**
  - This type of topology combines the benefits of different types of topologies in one topology.
  - Error detecting and troubleshooting are easy. ➢
- **Disadvantages:**
  - There is a change in the hardware to connect one topology with another.

(2) Find out application case of each topology.

1. **Bus Topology:**
   - **Applications:**
     - Traditional Ethernet networks used bus topology in the past, although it's less common today due to the prevalence of switched Ethernet.
     - It costs very less.
2. **Star Topology:**
   - **Applications:**
     - Star topology is often used in LANs within organizations to ensure easy connectivity and troubleshooting.
     - Bus topology can be used in small, simple, and cost-effective networks.
3. **Ring Topology:**
   - **Applications:**
     - Ring topology was commonly used in Token Ring networks, although these are less common today.
4. **Mesh Topology:**
   - **Applications:**
     - In wireless sensor networks, mesh topology can provide better coverage and reliability.
5. **Hybrid Topology:**
   - **Applications:**
     - Hybrid topologies are common in large enterprise networks where different departments or locations may have different network topology needs.
     - Organizations that require both scalability and fault tolerance may opt for hybrid topologies to balance these needs.

(3) Which topology you will implement in your campus and why?

In a full mesh topology, every device is connected to every other device, providing multiple paths for data transmission.
Advantages: High redundancy, fault tolerance, no single point of failure, excellent for critical applications.
Considerations: Requires a large number of connections and can be expensive to implement and manage in large networks.

References used by the students:

1. **Author name: Archana Rao Starting year:2006**

https://www.tutorialspoint.com

2. **Author name: Sandeep Jain**

Starting year:2008

https://www.geeksforgeeks.com

3. **Author name: Sonoo Jaiswal**

Starting year:2018

https://www.javatpoint.com

Rubric wise marks obtained:

Program	(Excellent)(4)	(Good)(3)	(Fair)(2)	(Beginning)(1)
Installation of tool	Tool installed Properly	Tool executes with a minor error	Tool executes with multiple minor (easily fixed error)	Tool does not execute (0-1)
Topology Implementation	Tool displays correct output with no errors	Output has minor errors	Output has multiple errors	Output is incorrect (0-1)
Analysis/Interpretation of output	Analysis and understanding are as per requirement	Analysis and understanding are not as per requirement	Multiple misunderstanding	Analysis and understanding are not proper
Documentation	Practical is well documented	Missing required comment	Missing two or more required comments	Most or all documentation missing (0-1)